Structural Non-Isolability in Knowledge-Based Robot Fault Diagnosis

Chenxuan Zhou * **Jie Liang** *

* *Faculty of Engineering and Information Technology, University of Technology Sydney, Ultimo, NSW 2007, Australia (e-mail: chenxuan.zhou@student.uts.edu.au, jie.liang@uts.edu.au).*

Abstract: Robot diagnosis layers often score the failure modes of an FMEA knowledge base against per-signal deviations. On the voraus-AD benchmark, two or three modes tie at the top in 514 of 755 anomalous cycles, always within one of four groups that declare the same joints and signal families. Top-1 accuracy ranges from 0.119 to 0.200 over tie-break orders; a prevalence prior reaches 0.187. We report the tied group and search for statistics that separate it. For the load-torque group they raise held-out within-group accuracy from 0.562 to 0.840. For the collision group, random splits adopt statistics that leave-one-setting-out validation rejects.

Keywords: Fault isolation and root cause analysis, Structural analysis methods, Diagnosability analysis, Knowledge-based diagnosis, Anomaly detection, Robotics.

1. INTRODUCTION

An anomaly detector on robot telemetry can tell that a cycle is abnormal, but not why. To name likely causes, a line-specific knowledge base is often built from a failure mode and effects analysis (FMEA). Each failure mode declares the joints it affects and the signal families in which it appears, a retrieval function scores the modes against the per-signal deviations of the cycle, and the maintenance technician receives a ranked list of candidate causes.

We built such a layer for the public voraus-AD benchmark (Brockmann et al., 2024) and asked which failure modes its list can isolate. The declarations of the knowledge base form a fault signature matrix in the sense of Blanke et al. (2016). In this matrix, nine of the 14 failure modes fall into four groups whose members declare the same joints and signal families. In three groups the columns are identical; in the fourth, one column contains the other two. The scoring function is continuous, yet in 514 of 755 anomalies two or three modes of one group receive the same top score. Which of them is reported then depends only on the order in which the modes are listed (Sec. 5.2).

These ties change how the layer should be evaluated. During development we compared top-1 accuracy with uniform chance. When fault frequencies are skewed, a more useful reference is a prevalence prior, which ranks the modes and groups by their frequency in the training data and gives the same ranking for every cycle. On voraus-AD this prior is better than the retrieval on top-3, top-5, mean reciprocal rank (MRR), and group accuracy under every tie-break order (Sec. 5).

The ties also raise a practical question: can statistics of the signals that are already recorded separate the members of a group? For each non-isolable group we search for such a statistic, which we call a *repair*. The answer depends on how the search is validated. The `setting` field of voraus-AD identifies a physical fault instantiation, for example one added weight or one placement of a collision object, and each instantiation is repeated for 4 to 20 cycles. A random split over cycles puts cycles of the same instantiation into both training and test data, and this changes which repair is adopted as well as the reported accuracy (Sec. 7). Leakage of this kind is known in machine-learning-based science (Kapoor and Narayanan, 2023) and in fault-diagnosis benchmarks (Hendriks et al., 2022); here it affects the inner validation that selects the repair.

Contributions. We analyse the knowledge base as a fault signature matrix and report non-isolable groups instead of breaking ties (Sec. 4). We give two references for a ranked candidate list on this benchmark, a prevalence prior and the exact range of top-1 accuracy over tie-break orders (Sec. 5). We search predefined statistics of the existing signals for one that separates a group, and report the result as a certificate with held-out evidence, controls, and a cost class (Sec. 6). We show that grouping the validation by fault instantiation changes which repair is adopted (Sec. 7), and we give the number of labels from which a flat classifier becomes more accurate than retrieval followed by a within-group classifier (Sec. 8).

2. FAULT SIGNATURES AND ISOLABILITY

Structural diagnosability analysis asks, before any data are seen, which faults a diagnosis system can at best separate, using the structure of the model equations (Blanke et al., 2016; Krysander et al., 2008). Its standard object is the fault signature matrix, whose rows are tests or residuals and whose columns are fault modes. Under the exoneration assumption, two modes are non-isolable exactly when their columns coincide. Without it, isolability is directional: a mode whose column is contained in another mode's column cannot be isolated from that mode (Krysander and Frisk, 2008; Cordier et al., 2004). The analysis has been applied to public process benchmarks (Düstegör et al.,

2006), extended to graded distinguishability under noise (Jung et al., 2015), and related to spectrum-based fault localisation in software (Travé-Massuyès and Wotawa, 2024).

When the signatures do not separate the faults, signature matching in fault detection and isolation (FDI) and consistency-based diagnosis in the artificial-intelligence (DX) tradition (Reiter, 1987; de Kleer and Williams, 1987) both return a set of candidates, and for a complete set of tests without exoneration the two sets coincide (Cordier et al., 2004; Travé-Massuyès, 2014). A ranked list that breaks ties reports one member of this set and drops the others. We therefore report the non-isolable group of the top-ranked mode, which contains every mode tied with it.

The usual remedy for a non-isolable group is sensor placement, which finds an additional measurement that makes the group isolable at the lowest cost (Krysander and Frisk, 2008; Frisk et al., 2009). Our case differs in two ways. There is no analytical model of the cell, and the rows of the signature matrix are declarations in an FMEA knowledge base (affected joints, signal families, cycle phase) rather than residuals of a model. What can be searched are statistics of the existing signals that the deployed representation does not use. This is test selection with data-driven evaluation, close to the hybrid scheme of Jung et al. (2018). A statistic that needs a new measurement corresponds to sensor placement; the others change only how the recorded signals are summarised. The evidence for a statistic is its accuracy on held-out fault instantiations.

3. SYSTEM AND DATA

Benchmark. voraus-AD (Brockmann et al., 2024) contains machine data from a six-axis robot performing a pick-and-place task, with 12 categories of physically induced anomalies. We use the 100 Hz release, which has 2122 cycles: 755 anomalous and 1367 normal. The official training data are 948 normal cycles, and the remaining 419 normal cycles are held out. The anomalous cycles fall under 72 `setting` values. Each value identifies one physical fault instantiation, repeated for 4 to 20 cycles (median 10), and three categories have a single instantiation.

Evidence record. We evaluate diagnosis on all 755 labelled anomalous cycles, which amounts to assuming perfect detection; in deployment an anomaly detector such as that of Brockmann et al. (2024) would select the cycles. Each of 26 channels (motor torque, torque sensor, joint position, and joint velocity on each of six joints, plus robot and system current) is resampled to 500 points over task progress and standardised against the 948 normal training cycles, $z = (x - \mu)/\max(\sigma, 0.05\,\mathrm{range}\,\mu)$. For the retrieval, the record of a cycle keeps the five channels with the largest peak $|z|$, the dominant joint (the joint with the largest peak torque deviation in N m), and the action phase at that peak.

Knowledge base and retrieval. The knowledge base is an FMEA encoded as a graph. Liao et al. (2025) automate the construction of such graphs for robots; ours was written by the authors from the voraus-AD anomaly descriptions. It declares 14 failure modes. Twelve map one to one to the benchmark's categories, and two, backlash and over-temperature, have no instances. Each mode declares its joint scope, its signal families, a cycle-phase hint, the direction and time course of its torque deviation, and a recommended check. The deployed score uses the first three. A mode m is scored against a record by $s = J + F + 0.2E + 0.2P + 0.15C$, where J is 1 if the dominant joint lies in the scope of m (0.5 for scope 'any'), F is the Jaccard overlap of signal families, E is the mean over the record's five channels of $\log(14/n)\log(1 + |z|)$, taken as 0 outside the families and joints of m, with n the number of modes that declare that family and joint, P is 1 if the record's phase agrees with the phase hint of m, and C is the fraction of the families of m that E counts. The form of the score and its weights were settled during development on these 755 cycles. When the three weights are chosen again on each training half of the protocol below, the choice differs on every split, and the test halves show no gain in MRR (0.362 against 0.366) or top-5 (0.600 against 0.651).

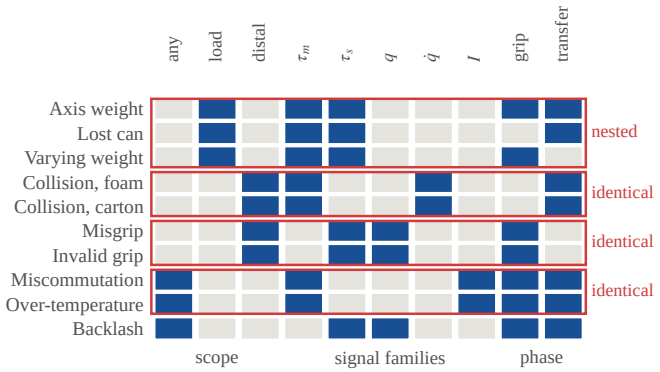


Fig. 1. Signature-matrix columns, drawn as rows, of the nine modes in non-isolable groups and of backlash, which has the gripping pair's families but not their scope. Scope 'load' is J1–J3, 'distal' J4–J6. Families: motor torque τ_m, torque sensor τ_s, joint position q, joint velocity $\dot{q}$, current I; a phase hint 'any' fills both phase cells.

Protocol. Unless stated otherwise, results are means over five leave-setting-out (grouped) splits. Each `setting` is assigned wholly to the training half or to the test half, stratified by category, so a learned component is always tested on instantiations it has not seen. Intervals are t-intervals over the five splits ($t_{0.975,4} = 2.776$). Because the splits re-partition the same 72 instantiations, the intervals describe the variation between splits on these data. Methods are compared split by split.

4. NON-ISOLABLE GROUPS

The fault signature matrix we analyse has one row per declaration that the score reads: joint scope, signal family, and phase. Fig. 1 shows the columns of the nine modes that share their joint scope and signal families with another mode. They form four groups: three load-torque modes (axis weight, lost can, varying weight), two distal collisions (foam, carton), two gripping faults (misgrip, invalid grip), and a commutation/over-temperature pair.

The score explains why these groups tie. Every term except phase agreement depends only on scope, families, and the record, so the members of a group can differ only in

the phase term. The score penalises extra families and the scope 'any', which corresponds to the exoneration assumption for these rows, but it gives the phase hint 'any' credit in every phase, grip (pick or place) and transfer. In the collision, gripping, and commutation groups the phase hints are also equal, so the columns are identical and the members always receive the same score. In the load-torque group the hints differ (axis weight: any phase; lost can: transfer; varying weight: grip), so the column of axis weight contains the other two. Neither payload mode can then be isolated from axis weight. Because a record has one phase, axis weight ties with one or both of them, and no member of the group ever ranks first alone.

The benchmark shows this pattern. In 514 of the 755 anomalies (68.1%) the top score is shared by two or three modes, and every tie lies inside one group. Of these ties, 498 are in the load-torque group: axis weight ties with lost can in 238 cycles and with varying weight in 199, and all three tie in 61 cycles whose peak lies outside the pick, place, and transfer phases. The other 16 are ties of the commutation pair. Setting the phase weight to any of seven values between 0 and 3 leaves the count at 514 and changes only which members tie. The collision and gripping groups never reach rank one. They are declared on the distal joints J4–J6, while the dominant joint, chosen by absolute torque deviation, is J1, J2, or J3 in every cycle.

These counts depend on how the knowledge base is written. If the phase of axis weight were declared as transfer instead of any, 315 ties would remain. The knowledge base also declares the direction and time course of each mode's torque deviation, which the score does not use and which differ within every group except the collision pair. Adding them to the score with weight 0.2 removes 513 of the 514 ties, but it lowers top-1 on all five splits and balanced recall (Sec. 5.1) on four of them, from 0.105 to 0.100. A list without ties is therefore not automatically more accurate.

5. TWO REFERENCES FOR THE RANKED LIST

5.1 The prevalence prior

Against uniform chance, which is 0.071 over 14 modes, the deployed retrieval has a top-1 lift of 2.37×. Uniform chance is a weak reference when categories differ in frequency, and on voraus-AD they differ widely: axis weight contributes 156 cycles and cable entanglement 10. The prevalence prior uses only these frequencies. It ranks the modes and groups by their frequency in the training half and gives the same ranking for every cycle.

Table 1 compares the prior with the retrieval. On top-3, top-5, MRR, and group accuracy the prior is ahead under every tie-break order of Sec. 5.2, and for top-3, MRR, and group accuracy it is ahead on all five splits. Top-1 is the exception, because it depends on the order. Under the declared order the retrieval is 0.018 below the prior and behind on four of five splits, while two of the 12 orders put it at or above the prior. Breaking ties by training-half frequency, which uses only the information the prior uses, gives 0.200, above the prior's 0.187. For group accuracy, always naming the most frequent group of the training half, which is load torque on every split, gives 0.401 against 0.342 for the retrieval.

Table 1. Retrieval and references on the five test halves. Group: accuracy of the predicted mode's group; for the prior, the most frequent group of the training half. Balanced: mean per-category recall, with tied modes sharing the credit; a constant scores $1/K$, with K the 10 or 11 categories in a test half. In parentheses: top-1 with tied modes sharing the credit. All orders: the retrieval's range over the 12 tie-break orders of Sec. 5.2. Two-stage: fixed {sign, step}, Sec. 6; flat classifier: Sec. 8.

	top-1	top-5	group	balanced
uniform chance	0.071	0.357	0.150	0.071
prevalence prior	0.187	0.705	0.401	0.095
retrieval	0.169 (0.160)	0.651	0.342	0.105
all orders	0.119–0.200	0.640–0.651	0.342	0.092–0.119
two-stage	0.309 (0.302)	—	0.342	0.208
flat classifier	0.525	—	0.574	0.397

Balanced recall, the mean of the per-category recalls, is $1/K$ for a constant predictor whatever the class frequencies, which averages 0.095 here, so on this measure the prior has no advantage. The retrieval's balanced recall also depends on the tie-break, and we let tied modes share the credit. The retrieval then reaches 0.105. This is above the constant on every split, by 0.002 to 0.016, and one tie-break order puts it below. Its expected recall is zero on six of the 12 categories, so the evidence in a cycle adds little to the frequencies. Comparisons between methods below use balanced recall in this form.

5.2 The tie-break interval

The scores do not determine which member of a tied set is reported; our implementation reports the member declared first. Ties involve only five modes, always within their group, so the $3! \cdot 2! = 12$ relative orders of these modes cover every way the result can change. Table 1 gives the range over all of them: top-1 accuracy runs from 0.119 to 0.200, a width of 0.082. The declared order gives 0.169, and the mean over the orders, 0.160, equals the top-1 accuracy obtained when tied modes share the credit (McSherry and Najork, 2008).

This spread is larger than the variation due to the choice of split. Over the orders the five-split mean has standard deviation 0.027 and range 0.082, while for the declared order the standard deviation over the five splits is 0.015 and the range 0.040. More splits would not narrow the first spread, because one order applies to all splits. Both ranges are larger than the 0.018 by which the retrieval is below the prior, so top-1 alone does not settle which of the two is better. An oracle that always picks the correct member of the tied set reaches 0.281. It bounds rules that choose inside the tied set, but not rules that choose inside the whole group, because the phase term has already removed the true mode from 46 tied sets.

The group of the top-ranked mode does not depend on the order, because no tie crosses a group: group accuracy is 0.342 under every tie-break rule. Taken as a set, the group contains the true mode in 34.2% of cycles, with 2.34 modes on average. The same number of modes taken from the

top of the prevalence list contains it in 38.5%. The group is ahead only on coverage averaged over categories (0.243 against 0.222). The knowledge base thus fixes how large the answer set is, but the group of the top-ranked mode is less often correct than a choice based on frequency alone.

6. TEST SELECTION OVER REPRESENTATION DIMENSIONS

Sec. 4 showed that the deployed score cannot separate the members of a group. We now ask whether other statistics of the existing signals can. For each non-isolable group we search for a repair, a set of such statistics with which a classifier, the *resolver*, separates the members.

Dimensions. The classifiers in this section read statistics of z. The retrieval uses only the peak magnitude $\max|z|$, and *mag* denotes this peak on each of the 26 channels. The candidate dimensions are further statistics of z: *sign* (mean of z), *step* (last third of the cycle minus first third), *payload* (carrying minus empty actions), *shape* (standard deviation of z), and *unread* (peak, signed, and payload statistics of two recorded channel families outside the 26). Each dimension has a cost class. It is *free* if it can be computed from the 26 channels, *pipeline* if it needs other recorded channels, and *hardware* if it needs a measurement the cell does not have. None of our dimensions is of hardware class.

Search and certificate. Every learned component is a standardised ℓ_2 logistic regression ($C = 1$). The search starts from mag and is greedy. At each step it adds the dimension whose gain in grouped inner cross-validation accuracy on the training half most exceeds 0.005 times its cost factor (1 for free, 6 for pipeline), and it stops when no gain does. The inner validation has $\min(4, n)$ folds, with n the fewest training settings of any member. The selected set is adopted on a split if its inner accuracy exceeds the share of the most frequent member in the training half by 0.05 and the accuracy of mag alone by 0.02, and the repair is adopted for the group if this holds on a majority of splits. If a member has fewer than two instantiations in the training half, no search can run; we then evaluate the set of all free dimensions and do not adopt it. The test half is used only for the certificate, which gives the dimensions, their cost class, the held-out accuracy within the group, and three controls. Table 2 lists the accuracies. Varying the cost factor, the minimum gain, and both margins does not change for which group a repair is adopted (repository).

Adopted. Within the load-torque group the repair reaches 0.840 on held-out instantiations. The deployed features reach 0.562, the majority class 0.504, and the same dimensions with permuted labels 0.431, and the repair is adopted on all five splits. The dimension selected most often is sign, on four of five splits (step on three, unread and shape on two each); mag with sign alone already reaches 0.827. Because unread enters on two splits, the worst-case cost class of the certificate is pipeline. The free set {sign, step}, however, reaches 0.869 against 0.872 for the best set with unread channels, so no additional channel is needed. With {sign, step}, most of the remaining errors are between the two payload modes: on cycles of these two modes the resolver is right in 0.80 of cases, against 0.93 and 0.98 for the pairs with axis weight. All these figures are computed on cycles whose true mode is in the group. They describe how well the repair separates the members once a cycle has been routed to the group; the routing itself is examined below.

Not adopted. Carton collision has three instantiations, so each training half contains one of them and no grouped search can run. The set of all free dimensions reaches 0.776 against a majority rate of 0.718. Each split trains on one carton instantiation and tests on the other two, and three of the five splits test on the same pair, so the five values are not independent. Sec. 7 compares the protocols on this group with folds on which the search can run.

Undetermined. Each gripping mode has one instantiation, so no grouped split can test a repair.

Untestable. voraus-AD contains no over-temperature fault, so no search is run. The recommended check for over-temperature names a statistic that would separate the pair: current against torque for the same motion.

We then place the resolver behind the retrieval. In this two-stage system, a cycle whose top group is load torque is passed to the resolver, and every other cycle keeps the retrieval's answer. With dimensions searched on each split, top-1 in declared order rises from 0.169 ± 0.018 to 0.302 ± 0.047. Fixing the dimensions selected on most splits, {sign, step}, gives 0.309 ± 0.031, a balanced recall of 0.208 (from 0.105), and 0.869 within the group. This choice was made with all five splits in view; choosing the dimensions from the other four splits each time gives 0.301 and 0.846, lower by 0.008 and 0.023. The gain of the fixed configuration over the retrieval is positive on every split, from 0.126 to 0.154.

The gain comes from the resolver, and the routing step costs accuracy. Sending every cycle to the load-torque resolver gives 0.349 and a balanced recall of 0.237, more than the two-stage system on all five splits. On the test halves the load-torque group is the true answer for 40.1% of cycles but is ranked first for 66.1%. The retrieval routes 69% of the load-torque cycles to the group, and 58% of the cycles it routes there are other faults. Of the routed cycles the resolver labels 0.37 correctly (0.88 of the load-torque cycles among them), against 0.16 for the retrieval alone. Sending every cycle to the resolver also reaches the 31% of load-torque cycles that the retrieval ranks elsewhere. The collision and gripping groups are never ranked first (Sec. 4), so in this system their repairs would never be applied.

7. GROUPED AND RANDOM VALIDATION

Each `setting` value is one physical fault instantiation. A random split over cycles therefore places cycles of the same instantiation in both halves, and a model can learn to recognise the instantiation instead of the fault. How much this matters depends on whether a method learns from labels. The flat classifier of Sec. 8 falls from 0.800 under a random split to 0.525 under leave-setting-out, while the retrieval, which learns nothing from the training half, gives 0.161 and 0.162 over 20 splits of each kind.

The type of split also changes which repair is adopted. For the load-torque group the repair is adopted under both protocols (Table 2), but for the collision group the protocols disagree. Under a random split the inner

Table 2. Repair certificates: accuracy within the group on held-out cycles of its members. Grouped: leave-setting-out, the protocol we report. Ungrouped: the same code under a random split over cycles, including its inner validation. Deployed: the same classifier on the mag features alone. Majority: accuracy of always naming the training-half majority class. Permuted: the selected dimensions refitted on training labels shuffled over cycles. Deployed, Majority, and Permuted are grouped. †One carton instantiation in each training half, so no grouped search can run; the value is for the set of all free dimensions.

Group	Ungrouped	Grouped	Deployed	Majority	Permuted	Verdict
load torque (3 modes)	0.933	0.840	0.562	0.504	0.431	**adopted** on 5/5 splits, both protocols
distal collision (2 modes)	0.830	0.776†	0.709	0.718	0.608	**not adopted**: search cannot run; adopted ungrouped
gripping (2 modes)	one instantiation per member					undetermined
commutation/temp (2 modes)	no over-temperature instance in voraus-AD					untestable

validation, whose folds share instantiations, reaches 0.85 to 0.96, and the repair is adopted on all five splits. On held-out cycles, however, it exceeds the majority rate of the same split by only 3 of 47 cycles (0.830 against 0.766). Because no grouped search can run under the 50/50 leave-setting-out split (Sec. 6), we compare the protocols with nine folds and the same search and rule. With folds stratified over cycles, the repair is adopted on 9 of 9 folds. With one setting held out per fold (leave-one-setting-out), it is adopted on 4 of 9 folds, and its pooled held-out accuracy is 0.67 to 0.74 over ten inner seeds, below the majority rate of 0.766 for every seed.

A permuted-label control with labels shuffled over cycles does not show this. It scores below the collision repair on all five splits (0.53 to 0.81 against 0.83), so the repair passes it. Labels shuffled over cycles are no longer tied to the instantiation, so the permuted model cannot gain from recognising instantiations, while the real model can. Permuting labels over whole settings keeps each label tied to an instantiation. Under the random split, 5 to 10 of the 84 possible labellings then reach the observed collision accuracy on each split ($p = 0.06$ to 0.12), so this control does not support the collision repair, whereas none of 300 labellings reaches the load-torque accuracy ($p < 0.004$). Learned comparisons on this benchmark should therefore group by setting in the inner validation as well as in the outer split.

8. CONTROLS AND LIMITS

8.1 The flat classifier

The resolver uses features that the retrieval ignores, so we also compared the whole system with a flat classifier on the same features. A flat 12-way logistic regression over all dimensions reaches 0.525 top-1, 0.397 balanced recall, and 0.574 group accuracy under the same protocol, against 0.309, 0.208, and 0.342 for the two-stage system. It recognises foam collision in 0.506 of cases, where the retrieval and the prior both score zero.

The knowledge base might still reduce the number of labels needed. We tested this over 20 paired leave-setting-out splits, training the classifier on k labelled cycles and the two-stage system on k labelled cycles per group. Only the load-torque resolver receives test cycles, so only its labels are used. Both systems are scored on the 12-way task. At $k = 3$ the two-stage system, which is then close to the knowledge base alone, has the higher balanced recall on 14 of 20 splits. From $k = 8$ the classifier has the higher top-1 (0.225 against 0.208, 15 of 20 splits) and balanced recall (0.154 against 0.136, 15 of 20), and at $k = 320$ it reaches 0.550 against 0.304. If the two-stage system receives the same labelled cycles as the classifier, it is behind at $k = 8$ on all 20 splits. Once eight or more labelled cycles are available, the flat classifier is the better choice.

8.2 What would settle the open verdicts

Two verdicts remain open. Gripping would need at least two instantiations per mode for any grouped protocol, and voraus-AD has one. For collision the limit is our split. A 50/50 grouped split leaves $\lfloor N/2 \rfloor$ instantiations per mode for training, the grouped inner search needs two, and carton has $N = 3$. Under leave-one-setting-out the search can run, and the repair is still not adopted (Sec. 7). We also ran sweeps over N on injected and on simulated faults (repository), but in both the deployed features already separate the pair, so they do not tell how many instantiations a real non-isolable pair needs.

9. CONCLUSION

On voraus-AD, the ranked list of an FMEA-based diagnosis layer is limited by the structure of its knowledge base: 68.1% of anomalies tie at rank one, almost all in the load-torque group. A single top-1 figure then depends more on the tie-break order than on the split, and a prevalence prior is better on top-3, top-5, MRR, and group accuracy under every order.

Reporting the non-isolable group makes this limit visible, and a search over statistics of the existing signals can then try to separate the group, with held-out evidence, controls, and a cost class. One of the three groups with benchmark instances is separated on unseen instantiations, mainly by the sign of the deviation and at no extra cost. The search has to be validated by instantiation: under a random split it adopts a collision repair that leave-one-setting-out validation does not, and a permuted-label control over cycles passes it.

Isolability can be checked before any data are seen, but removing ties does not by itself make the list more accurate: scoring the declared direction and time course

removes them and lowers accuracy. With eight or more labelled cycles, a flat classifier over the same features is more accurate than the two-stage system. A change of representation should therefore be adopted on evidence from unseen instantiations. All results here come from one benchmark and one knowledge base written by the authors; the same checks can be run on any FMEA-based diagnosis layer before it is deployed.

Code and data. Evidence records, certificates, all analyses and the producing scripts are archived at `https://doi.org/10.5281/zenodo.22089173`.

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